

Proactive Shield Final Presentation

Team 09

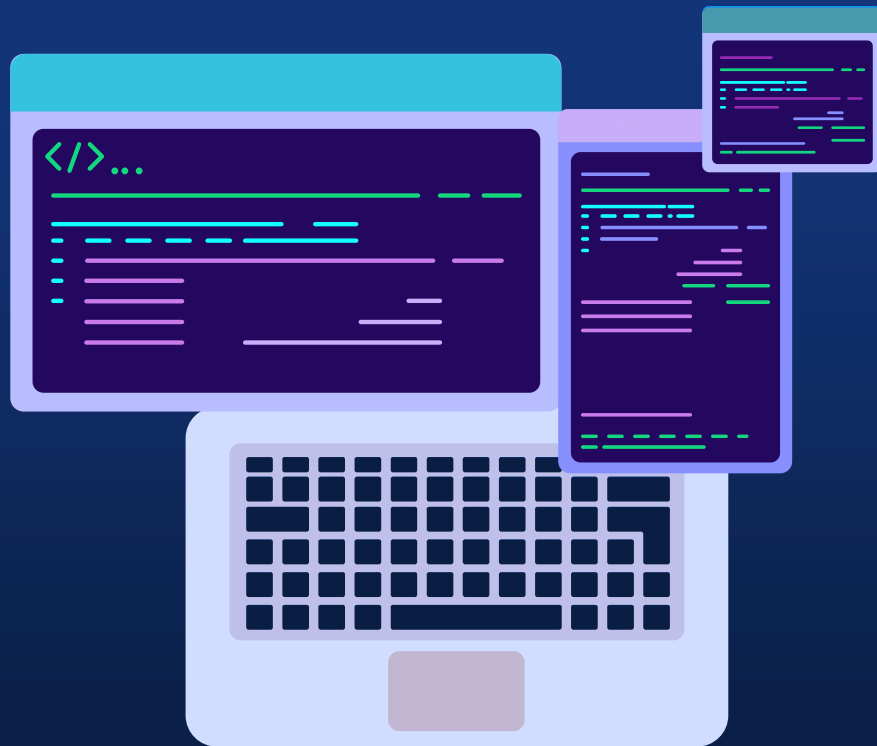
Member

Xuanhe Yang

Jingxiao Han

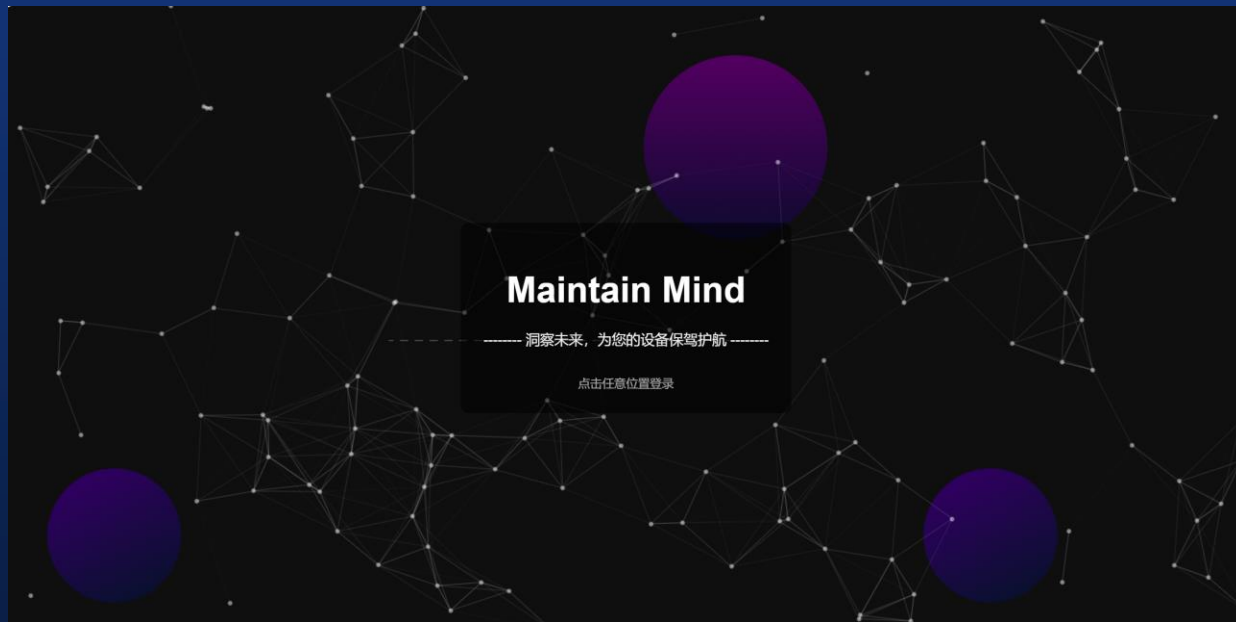
Tong Li

Fubin Chen

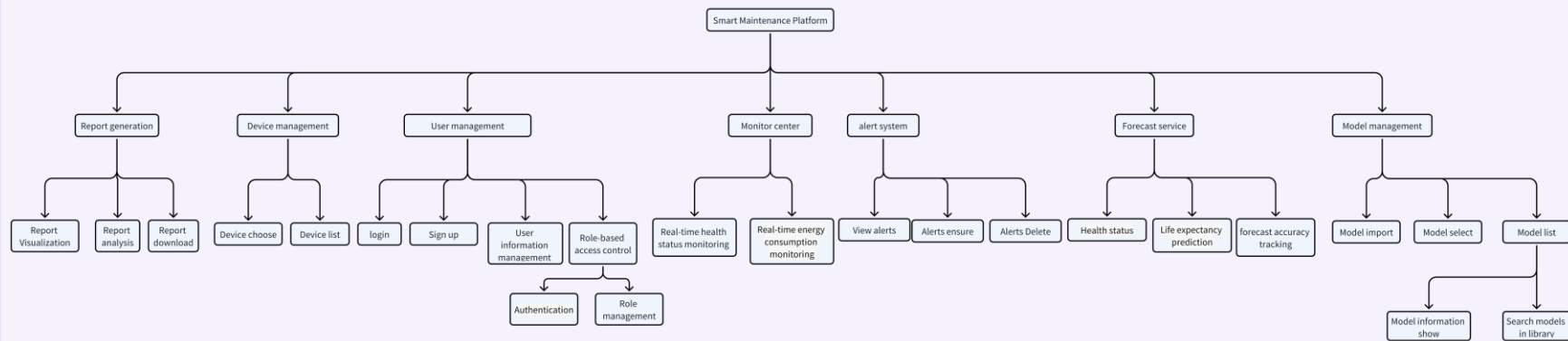


Project Overview

- Proactive Shield



- A device health assessment website based on AI models



Device Center

User Part

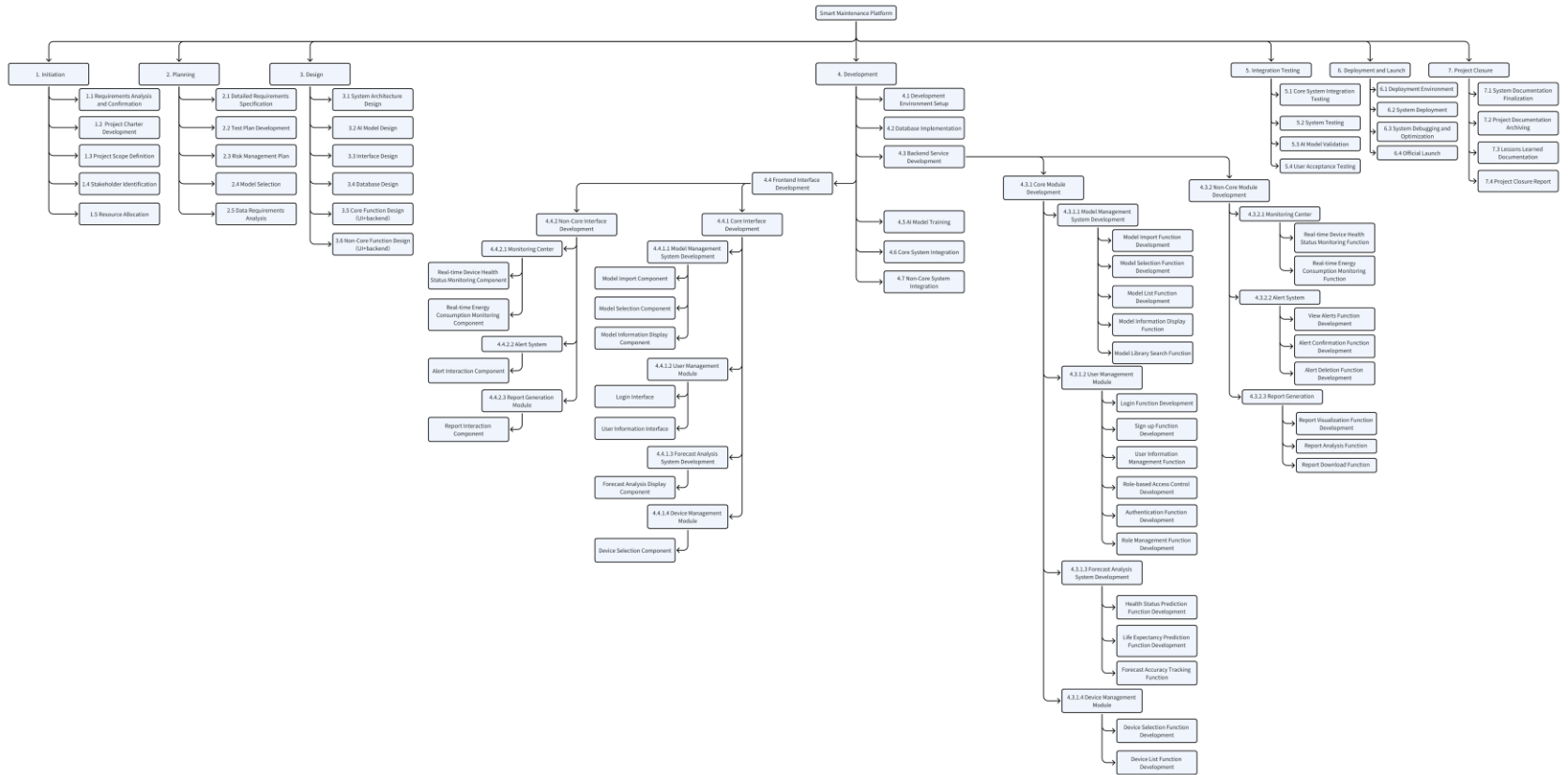
Monitor Center

Forecast Service

Alert System

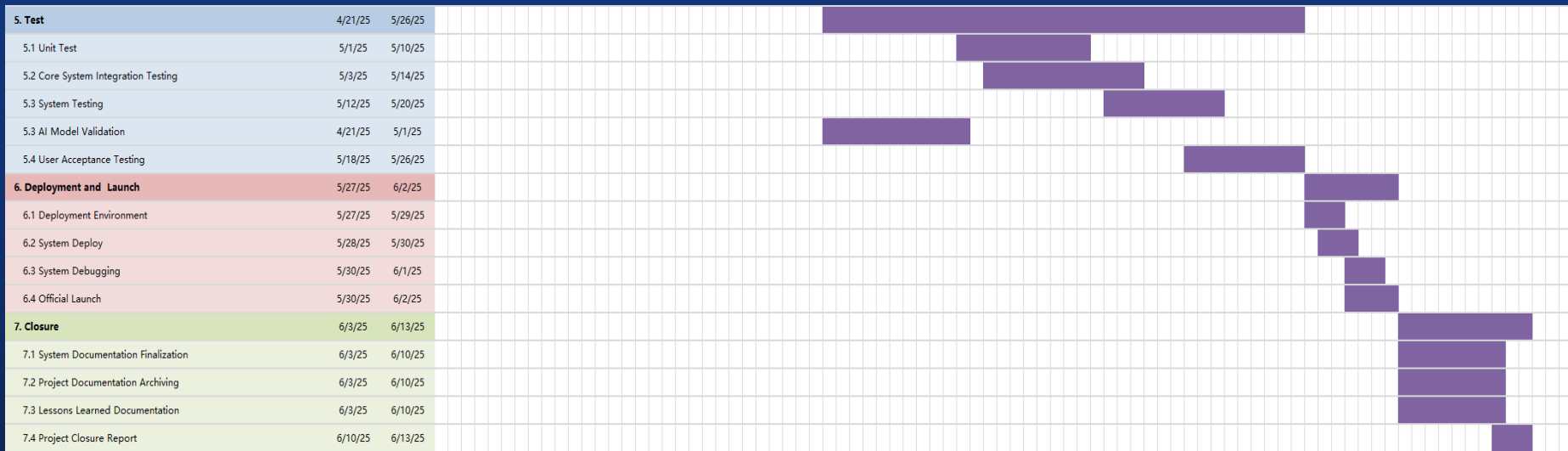
Report System

WBS





Gantt Chart of the project



Following the teacher's advice, we redrew the Gantt chart with a higher degree of parallelism

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01

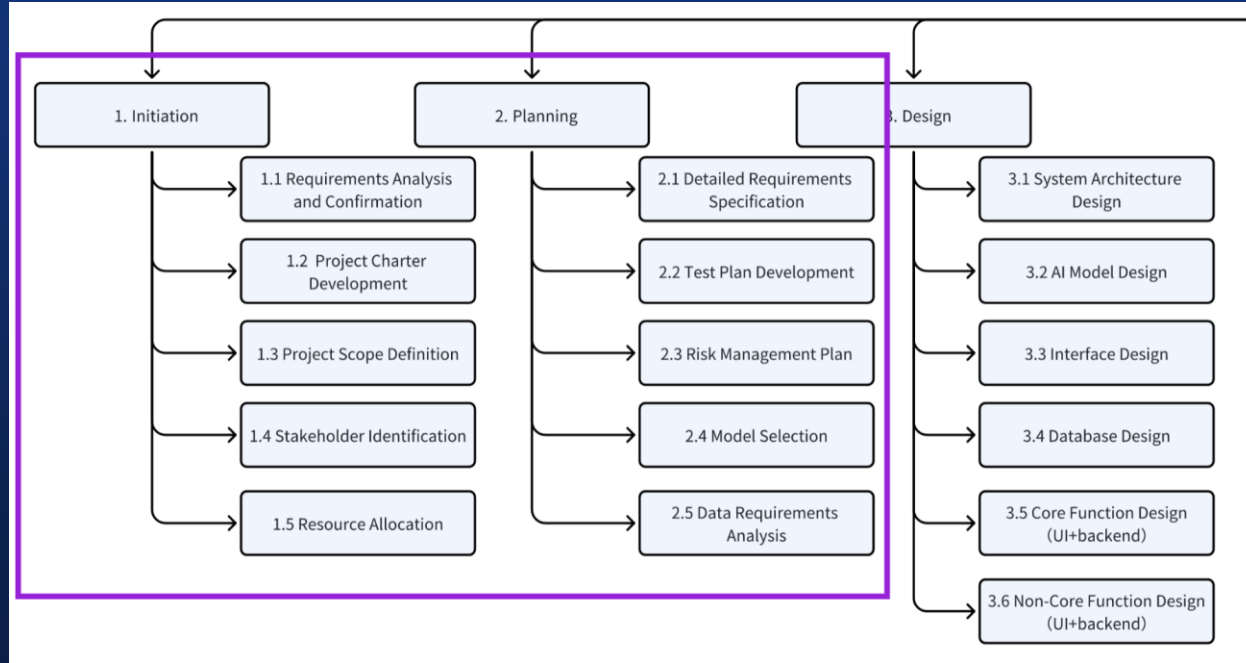
Summary of biweekly Report && Milestone





Summary of Biweekly Report

• The First Biweekly Report (3.24 - 4.6)

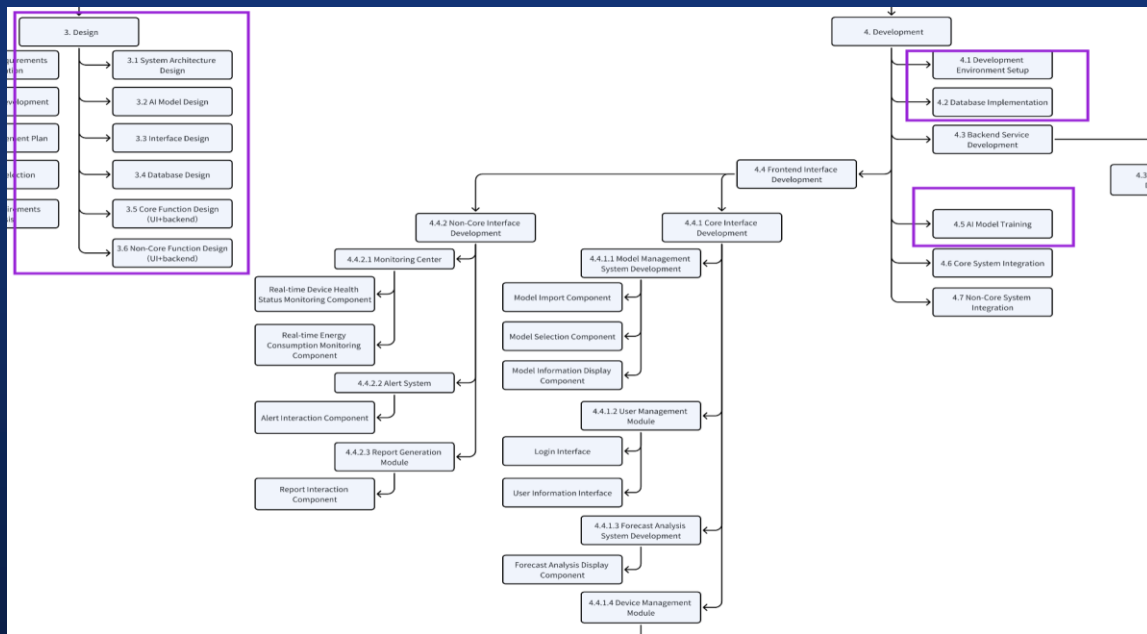


Overall, we have completed the **Initiation and Planning Part**



Summary of Biweekly Report

• The Second Biweekly Report (4.7 - 4.20)



We are about to complete the Design part and start the Development part

Summary of the First MileStone Meeting

● The First MileStone Meeting



Group 9 Milestone 1 Review Meeting

杨烜赫

Group 9 Milestone 1 Review Meeting

Date: 2025/4/19

Time: 18: 00 - 18: 45

Place: online

Agenda: 1. Project Introduction
2. Project Progress Report
3. Future Plans

Tencent Meeting: 112-573-166

<https://meeting.tencent.com/dm/hplTXUnilPjB>

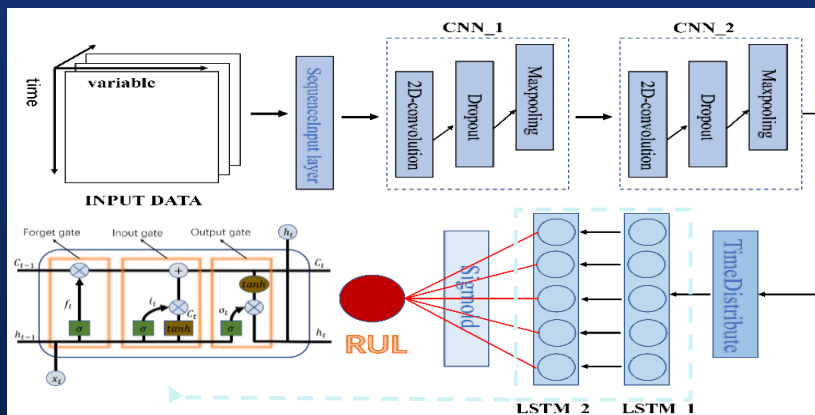
Note: After the meeting, please reply with your review comments in the discussion area. Thank you!

First Biweekly Report--Team 09.pdf

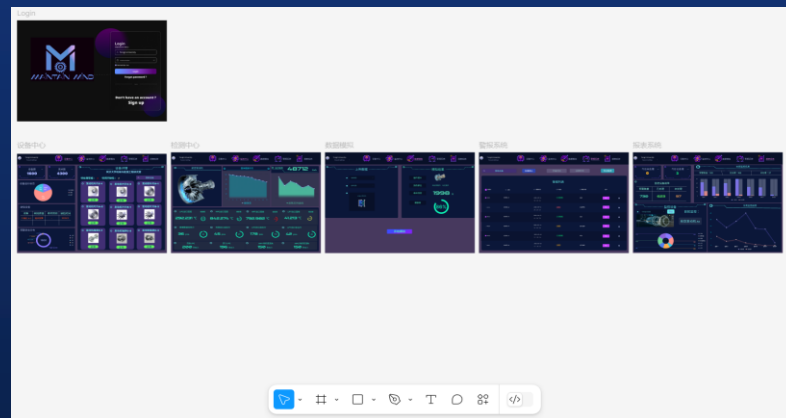
Summary of the First MileStone Meeting

- The First MileStone Meeting (First Biweekly + Second Biweekly)

AI model Design



Interface Design



We presented the **important achievements (Design Part)** of this project at the meeting

Summary of the First MileStone Meeting

● The First MileStone Meeting Review



黄俊哲
Sunday

you have successfully completed both the Initiation and Planning phases of the Smart Maintenance Platform project on schedule—finalizing the Project Charter, detailed requirements, risk management plan, testing strategy, and system architecture—and setting up the development environment and collaboration tools; with all WBS tasks for initiation, planning, and architecture design 100% complete, the team is now poised to begin module development (Device Center, Monitoring Center), build the sensor data preprocessing pipeline, implement the initial fault-detection model, design the database schema, and kick off frontend prototyping.

before diving into module development, set up a lightweight continuous integration workflow that runs automated tests on your data-preprocessing pipeline and initial service stubs—this will catch integration issues early and keep your architecture stable as you build out the Device and Monitoring Centers.

Reply



陈沛锦
Apr 19, 2025

The report of Team09 is detailed in content, with a clear Work Breakdown Structure (WBS), standardized and clear documentation, reasonable division of labor, and good completion status.

In the subsequent development phase, it is recommended that the team consider clear definitions of module interfaces and data flows to avoid potential dependency issues that may arise during the integration phase in advance. At the same time, it is recommended to further improve task division, optimize working hour calculation, and add consideration for cross module task collaboration in the work plan to ensure smoother and more efficient collaboration.

Reply



何升阳
Monday

我认为项目管理比较好，需要注意的是机器学习这个核心功能，在测试方面需要更多的体现，比如机器学习预测的准确性等等

Reply



刘文美
Apr 19, 2025

The introduction is very detailed, with clear information about what each person has completed and the timeline. The next steps are also well outlined. My suggestion is to explicitly mention the comparison between the planned hours and the actual hours for each task.

Reply



曹舒涵
Apr 19, 2025

- Overall, the work is progressing well, with the design documentation, testing documentation, and user manual all being updated in parallel.
1. I feel that the design document could benefit from a more detailed requirements analysis section. Using diagrams like use case diagrams to showcase the system's functionalities would make it clearer, rather than relying solely on text.
 2. Regarding the testing documentation, it could be more specific. For instance, adding concrete test cases would be helpful. Currently, the document still contains a lot of conceptual content, and it's not very clear how exactly the testing is being carried out.
(I've already received feedback on points 1 and 2 — the reason is that the documents are updated biweekly, so the latest content was not yet included in this round's version.)
 3. In the meeting, the sections on project background, resources, and risks were covered extensively. It would be better to condense this part slightly, as it felt a bit too lengthy.
 4. While the introduction of individual contributions was included, the presentation lacked more detailed introduction. It would be beneficial to include more specifics, such as which aspects of the WBS were implemented during this phase, the percentage of the finished part to check whether the project's current progress aligns with the expected timeline.

Reply



康文杰
Sunday

我认为您的团队所做的工作非常出色。项目主题真实创新，具有较强的实用性。你在这四个星期的工作也非常有条理和合理。系统开发结果符合您的预期。数据库和后端代码都相当完整，前端接口非常实用。您的文档也很严格。

我的建议是，如果你能把开发过程中遇到的问题、你是如何解决的、你如何调整开发计划和时间分配的细节添加进来的，这将有助于我更清楚地了解你的项目进度和细节。

Reply



关咏霖
Apr 19, 2025

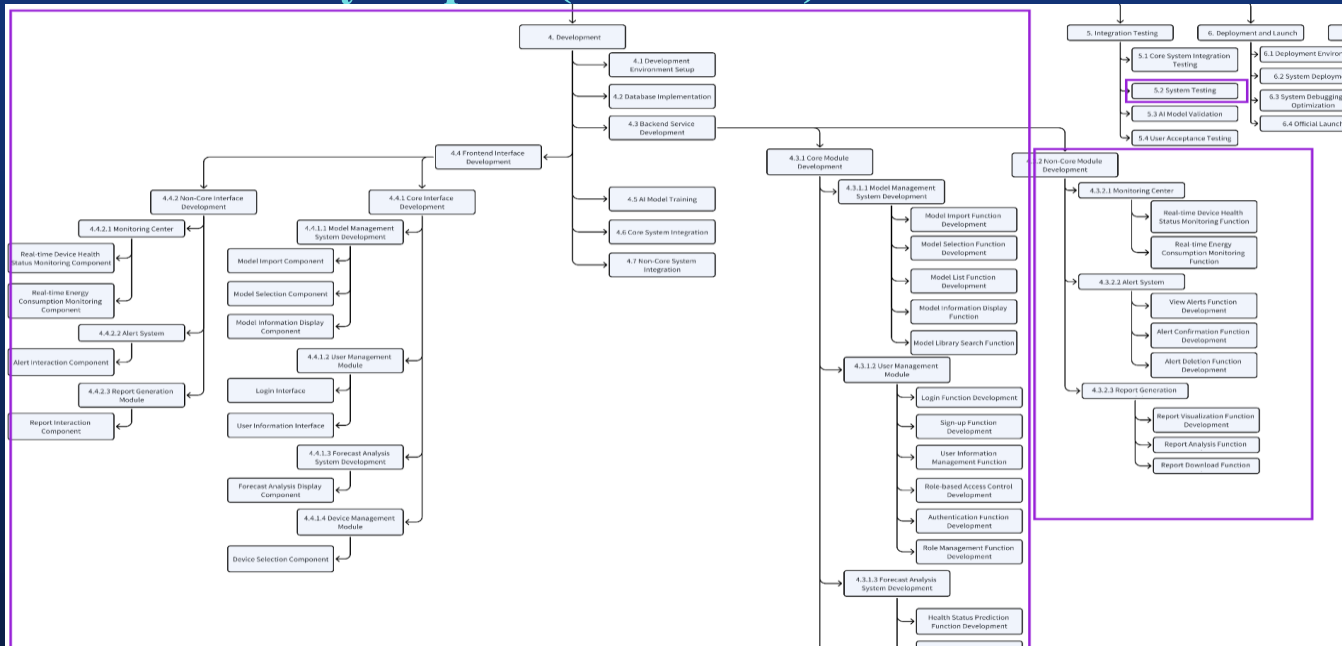
- 👉 Sending documents in advance for easy inspection by reviewers during the review process, with a complete review structure:
1. All tasks in the group division of work were assigned to a single person, without mentioning actual cooperation review meetings;
 2. The interface in the design document is not provided, but the user manual has been provided, indicating that the page elements have been determined and the document needs to be updated;
 3. Adding architectural design graphs to the design document would be more intuitive 📄

Reply



Summary of Biweekly Report

• The Third Biweekly Report (4.21 - 5.11)



We are about to complete the Development part and start the Testing part



Summary of the Second MileStone Meeting

● The Second MileStone Meeting



Group 9 Milestone 2 Review Meeting

杨烜赫

Group 9 Milestone 2 Review Meeting

Date: 2025/5/17

Time: 19: 00 - 19: 45

Place: online

Agenda: 1. Project Introduction

2. Project Progress Report

3. Future Plans

Tencent Meeting: 259-219-240

<https://meeting.tencent.com/dm/1V4yAloY3eJy>

Note: After the meeting, please reply with your review comments in the discussion area. Thank you!

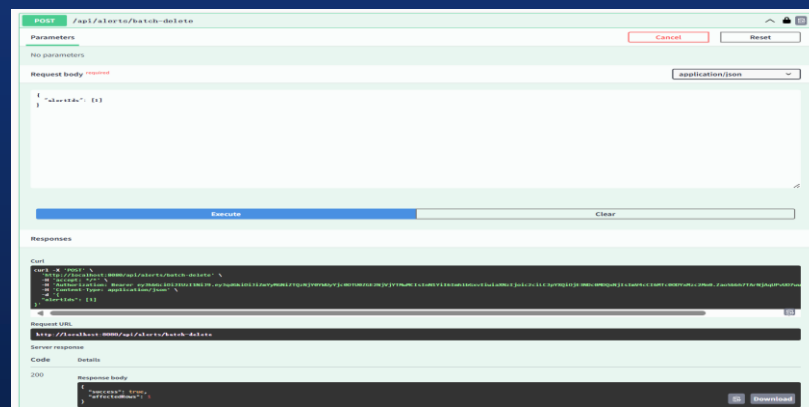
Summary of the Second MileStone Meeting

• The Second MileStone Meeting (Third Biweekly)

Frontend Development



Backend Development



We presented the important achievements (Development Part) of this project at the meeting

Summary of the Second MileStone Meeting

● The Second MileStone Meeting Review

 **黄俊哲**
Yesterday

It can be seen that your project has completed the core function development and AI model training verification. The core modules (equipment management, real-time monitoring, alarm system) and machine learning models (Skip-GANomaly anomaly detection accuracy>95%, CNN-LSTM life prediction RMSE<10 hours) have reached the design goals. The system integration test has passed and entered the final testing stage. I hope you can show a comparison between the current progress of the project (including the completion status of the members) and the expected progress in your report, so that we can see the completion status of the project more intuitively.

↩ Reply

 **陈沛峰**
Saturday

The team has made good progress by this milestone. Most of the main features are already developed, and the frontend and backend are working together well. The AI models have been trained and are now being connected to the system. Overall, things are moving faster than planned, and the work so far looks solid.

For the next step, it would be good to finish connecting the AI parts to the rest of the system and keep working on more system testing.

↩ Reply

 **曹舒通**
Saturday

The project is very complete, and the documentation is being updated synchronously. The structure of the code is clear and comments are also clearly written, and there are parts for code reuse. Here are some of my questions and suggestions:

1. In milestone2, there is a big difference in the working hours of the members, was it planned at the beginning of the project or did some accidents in milestone2 happened?
2. How exactly does the test work? The test documentation is a bit unintelligible.
3. At present, the progress of the project is not visually displayed, and I hope that it can be shown by the number of tasks completed or the Gantt chart.

↩ Reply

 **康文杰**
Yesterday

您的项目执行得相当不错。您的实施总体上遵循计划，开发稳步进行，并且已经建立了相对完整的原型系统。我的建议如下：首先，希望你能展示出每个团队成员的具体任务和时间表，这将有助于看到团队内部的分工。其次，我希望您在演示时将解释与甘特图对应起来，这样才能更直观地理解开发进度。

↩ Reply

 **刘文美**
Yesterday

The functionality is well implemented, and the code is well-structured with proper comments. However, the progress is not very clear. I'd like to ask whether your progress is ahead of schedule, behind schedule, or on track?

↩ Reply

 **关咏臻**
Saturday

- 1.What is the current unit test coverage status? Have all components been covered as planned?
- 2.During the code review, is the system fully integrated across frontend, backend, and database? Are the APIs already debugged and working?
- 3.In the second biweekly report, why did two members work on the same WBS task (WBS 3.3) ? How was the work divided?

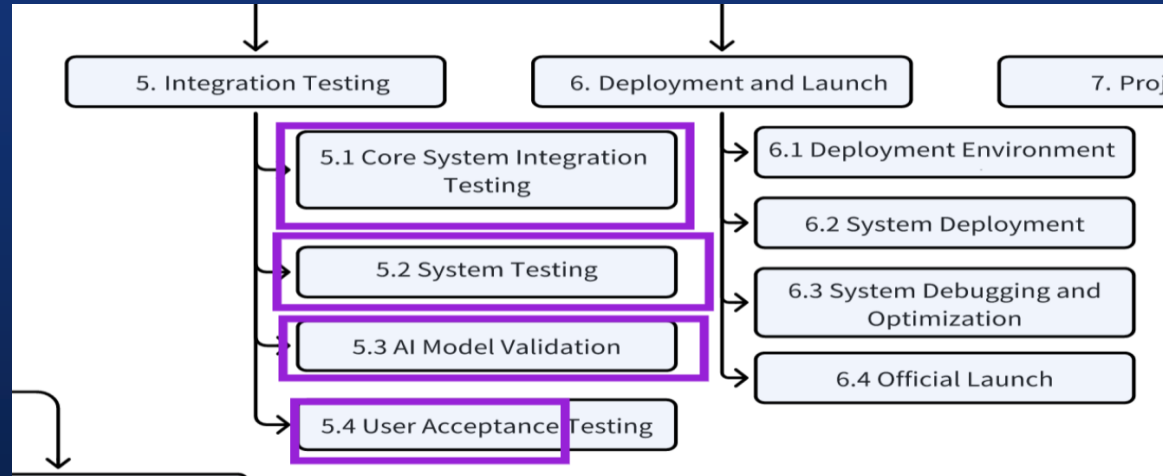
↩ Reply

 **何升阳**
May 25, 2025

截至当前里程碑，团队已取得显著进展。核心功能模块基本开发完毕，前端与后端的集成效果理想。人工智能模型已完成训练阶段，正逐步接入系统进行联调。**整体开发节奏超前于原计划**，已落地的工作成果稳定性良好。**建议：将系统全链路打通，尤其是AI与系统其他部分**

Summary of Biweekly Report

- The Fourth Biweekly Report (5.12 - 5.25)

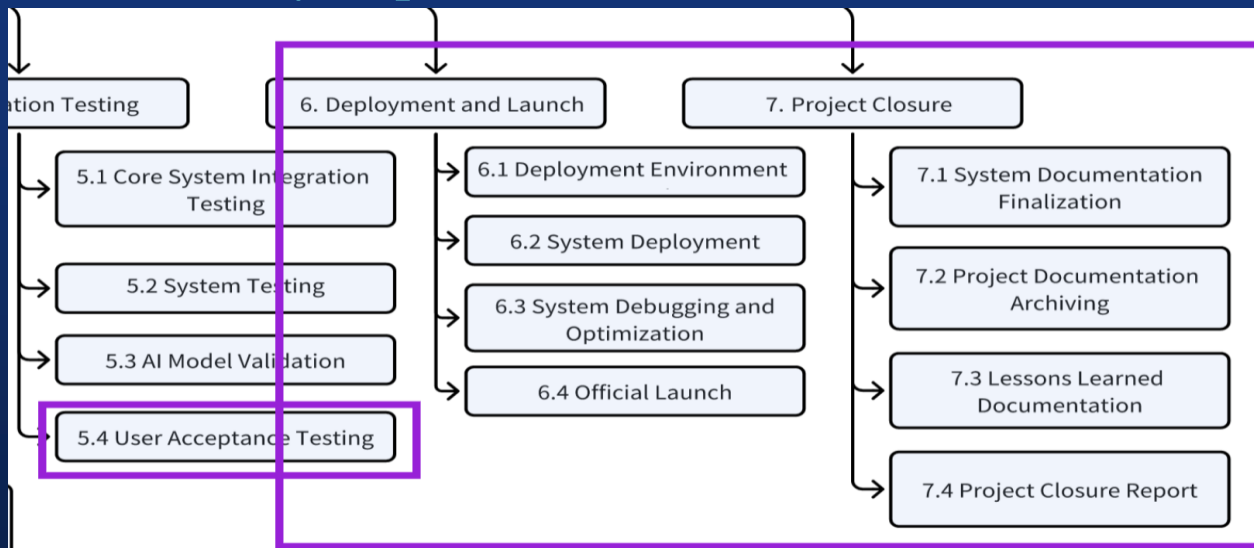


We are complete the Development part and about to complete the Testing part



Summary of Biweekly Report

• The Fifth Biweekly Report (5.26 - 6.15)



We are complete Testing, Deployment, Closure Part.

Our project was completed on time!



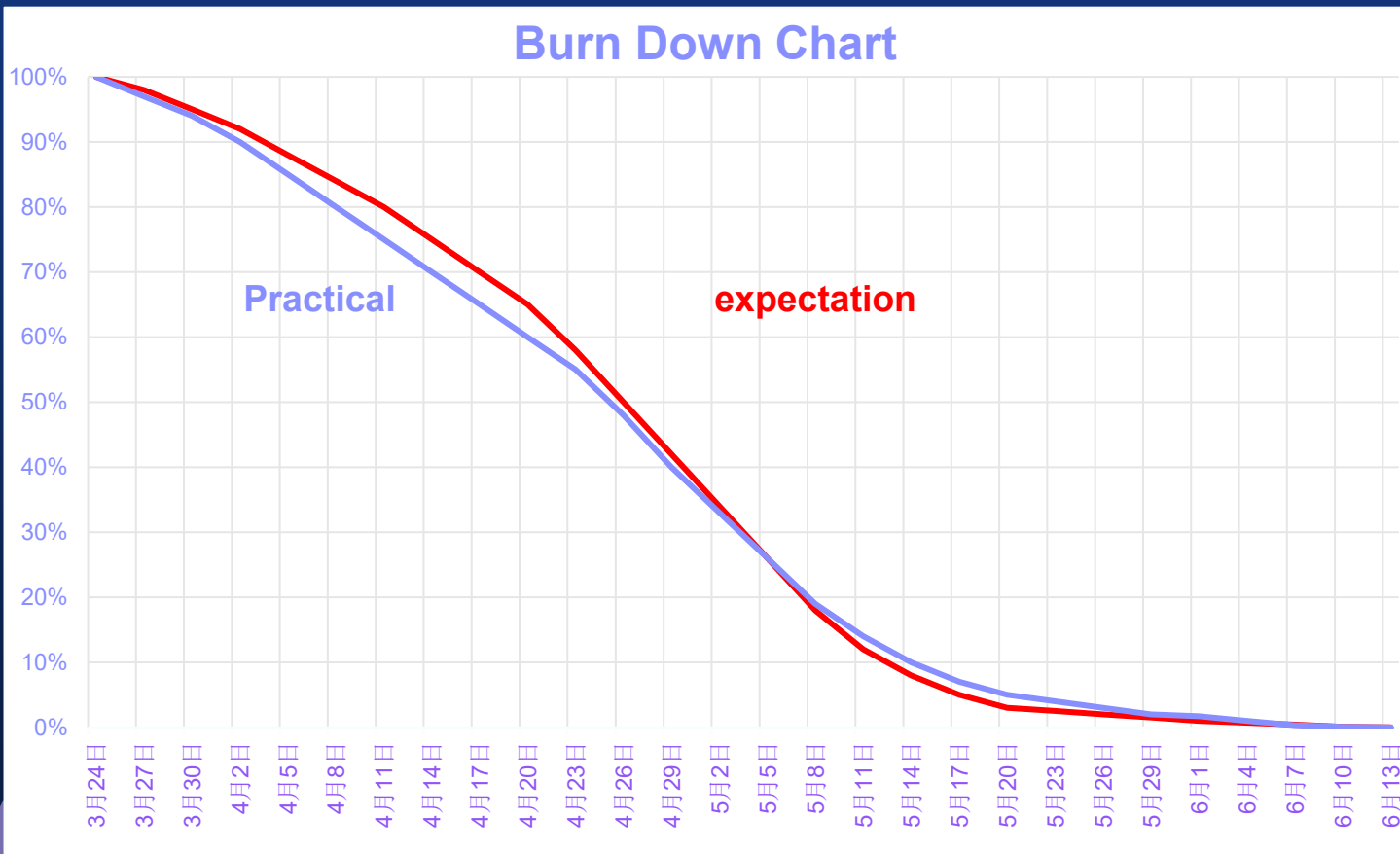


02

Current status of the course project

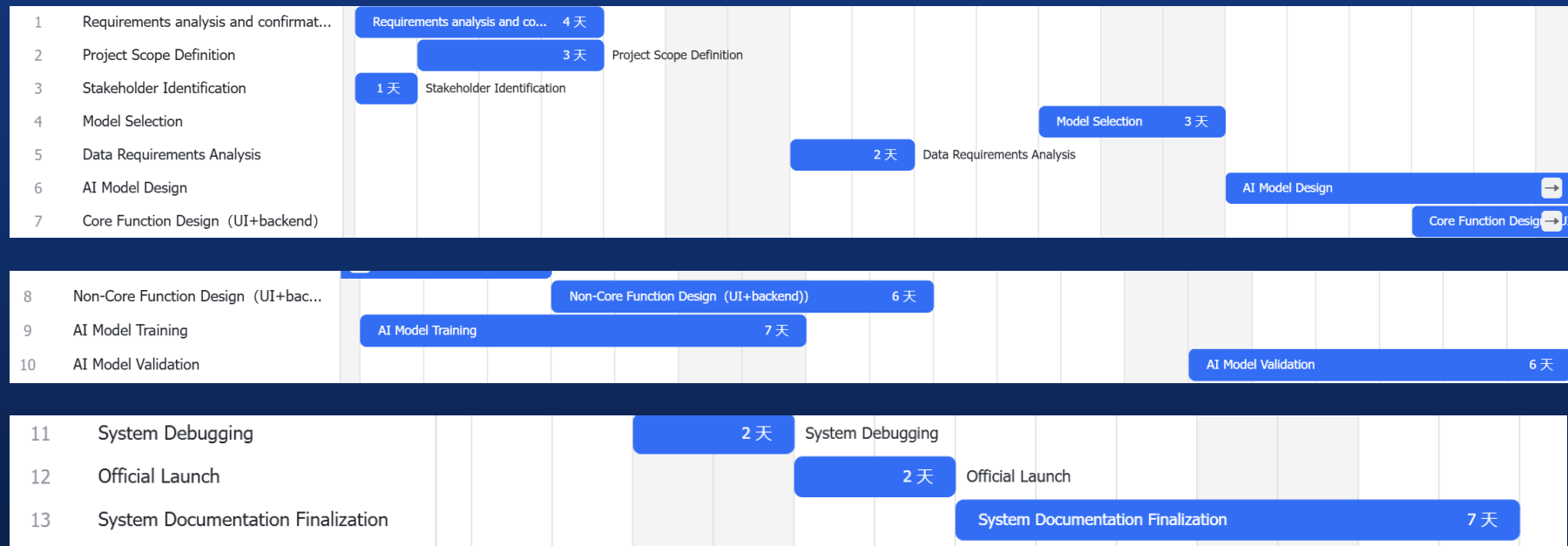


Burn Down Chart of the project



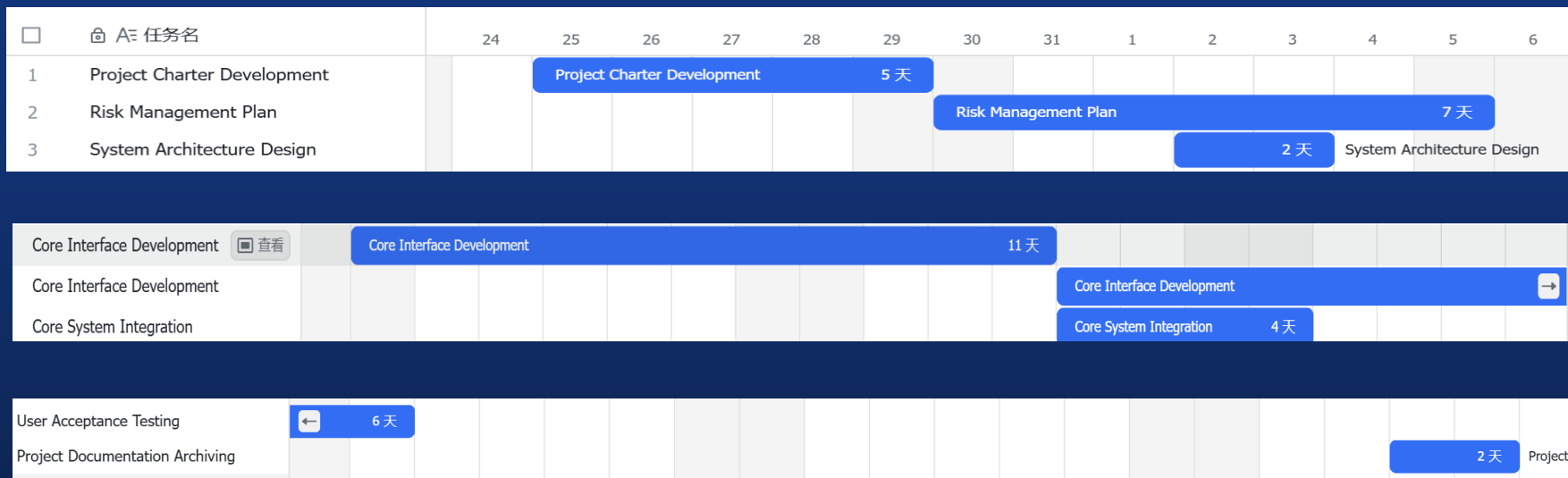
- Slope is larger in the middle for (Design Develop Test)
- The actual work is faster because well-prepared in advance
- We encountered problems during the testing stage which led to a slight delay

Gantt Chart of YXH



The member completed the task on time according to the Gantt chart

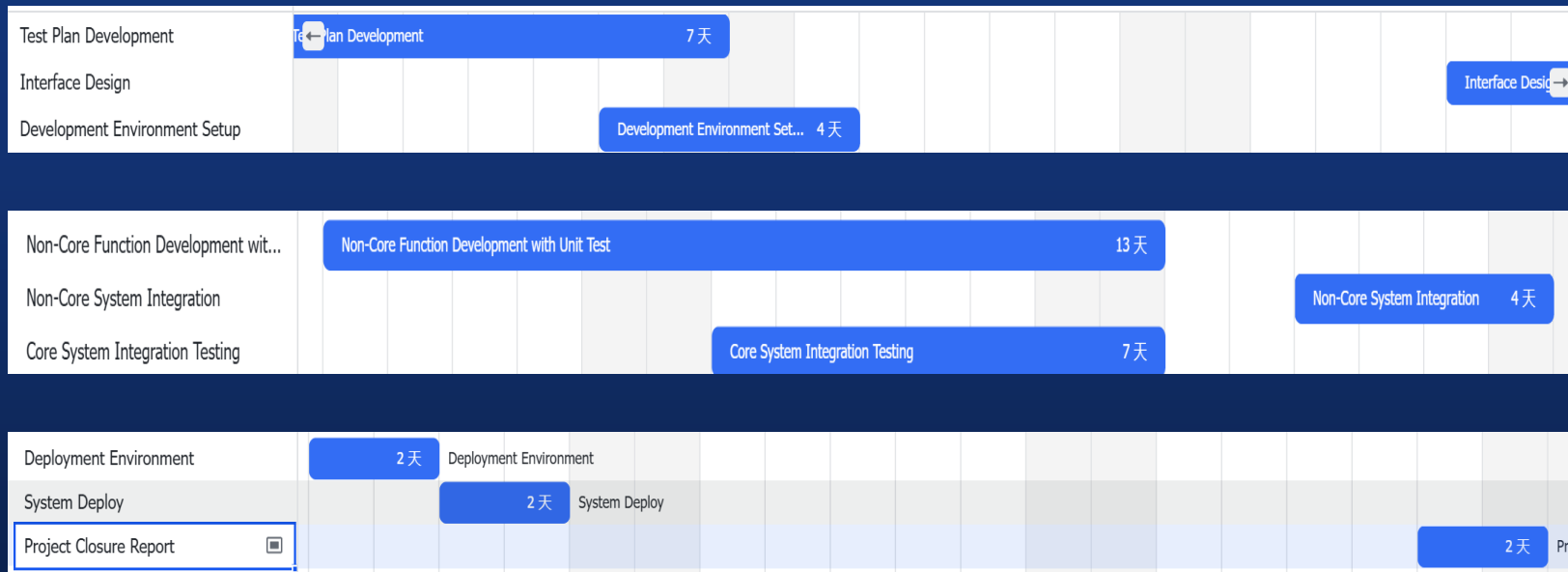
Gantt Chart of CFB



The member completed the task on time according to the Gantt chart



Gantt Chart of HJX



The member completed the task on time according to the Gantt chart

Gantt Chart of LT

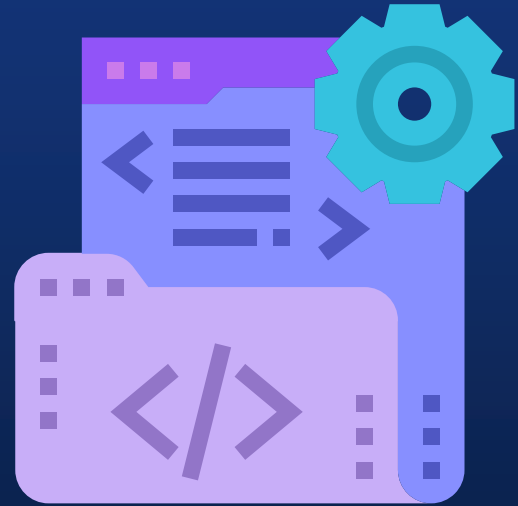


The member completed the task on time according to the Gantt chart



03

Github submission history



Github Submission Summary

● Submission Summary (Main Baunch)

Contributors

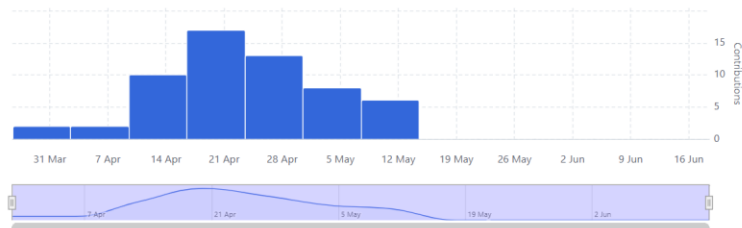
Contributions per week to main, excluding merge commits

Period: All

Contributions: Commits

Commits over time

Weekly from Mar 30, 2025 to Jun 15, 2025



PuBingA

22 commits 5,886 ++ 432 --

#1



Eurekahah

15 commits 17,862 ++ 252 --

#2



bambooshoot2

13 commits 3,283 ++ 386 --

#3



Augensternt

8 commits 20,617 ++ 74 --

#4



Commits

main

All users

All time

Commits on Jun 15, 2025

Merge remote-tracking branch 'origin/dev-lt'

Eurekahah committed 4 minutes ago

af93f26



Merge remote-tracking branch 'origin/dev-lt' into dev-main

Eurekahah committed 1 hour ago

b0f57be



Commits on May 17, 2025

修正部分代码bug

bambooshoot2 committed last month

be8d9f6



Commits on May 16, 2025

修复模拟模块测试

Eurekahah committed on May 16

b0e8484



Commits on May 15, 2025

更新部分代码bug

bambooshoot2 committed on May 15

27750b1



测试数据边界条件处理优化

Eurekahah committed on May 15

fef4999



Commits on May 12, 2025

集成非核心功能模块API和数据流

PuBingA committed on May 12

48d695f



Merge remote-tracking branch 'origin/dev-yxh' into dev-main

Eurekahah committed on May 12

b5948de



Commits on May 11, 2025

侧出报表函数还在进行

PuBingA committed on May 11

3d5fccd



Commits on May 10, 2025

Github Submission Summary

● YXH Submission Overview

Commits		
dev-yxh	All users	All time
Commits on May 4, 2025		
更新AI与后端的连接部分 Augenstern committed on May 4	f4cf5fb	
Commits on May 3, 2025		
继续更新测试脚本 Augenstern committed on May 3	294f3dd	
Commits on May 1, 2025		
更新测试脚本 Augenstern committed on May 1	1486a22	
Commits on Apr 27, 2025		
更新训练好的模型 Augenstern committed on Apr 27	9c37519	
Commits on Apr 26, 2025		
增加CNN-LSTM模型 Augenstern committed on Apr 26	0ee3241	
Commits on Apr 25, 2025		
更新模型 Augenstern committed on Apr 25	48b7354	
Commits on Apr 16, 2025		
更新模型train部分 Augenstern committed on Apr 16	cbde437	
初始化Almain	61f19e4	

Github Submission Summary

● CFB Submission Overview

Commits		
dev-cfb	All users	All time
Commits on May 12, 2025	集成非核心功能模块API和数据流 PuBingA committed on May 12 48d695f	
Commits on May 11, 2025	倒出报表函数还在进行 PuBingA committed on May 11 3d5fcdd	
Commits on May 10, 2025	报表系统实现关键函数 PuBingA committed on May 10 b2bec99	
Commits on May 9, 2025	报表系统功能函数补充中 PuBingA committed on May 9 300c27e	
Commits on May 8, 2025	报表系统格式优化中 PuBingA committed on May 8 4c2d187	
Commits on May 7, 2025	报表系统框架构建 PuBingA committed on May 7 9fa0a0b	
Commits on May 6, 2025	警报系统函数修复完毕 PuBingA committed on May 6 cc43552	

Github Submission Summary

● HJX Submission Overview

Commits		
dev-hjx		
All users All time		
Commits on Apr 28, 2025		
完成报表系统	Eurekahah committed on Apr 28	474d66e
Commits on Apr 27, 2025		
更新报表系统	Eurekahah committed on Apr 27	eef565e
Commits on Apr 25, 2025		
修改监控系统	Eurekahah committed on Apr 25	448336a
Commits on Apr 23, 2025		
新增监控系统	Eurekahah committed on Apr 23	98955bc
Commits on Apr 18, 2025		
注释部分错误实现	Eurekahah committed on Apr 18	878c960
更新警报中心	Eurekahah committed on Apr 18	7d7ac21
Commits on Apr 16, 2025		
开发警告系统	Eurekahah committed on Apr 16	af4c7fb
Commits on Apr 15, 2025		
初始化非核心功能基本框架	Eurekahah committed on Apr 15	3477d1c

Github Submission Summary

● LT Submission Overview

Commits		
dev-It	All users	All time
Commits on May 17, 2025		
修正部分代码bug	bambooshoot2 committed last month	be8d9f6
Commits on May 15, 2025		
更新部分代码bug	bambooshoot2 committed on May 15	27758b1
Commits on Apr 26, 2025		
添加演示图片	bambooshoot2 committed on Apr 26	aa5555d
Commits on Apr 25, 2025		
更新配置	bambooshoot2 committed on Apr 25	335c9c4
设备模拟与数据	bambooshoot2 committed on Apr 25	a8d1292
Commits on Apr 24, 2025		
设备管理	bambooshoot2 committed on Apr 24	8d56e82
Commits on Apr 23, 2025		
添加用户安全访问	bambooshoot2 committed on Apr 23	e4b1963
Commits on Apr 22, 2025		

More Detail See URL:<https://github.com/Augensternst/TJ-Software-management/commits/main/>

04

Quantitative statistical result



Tasks Statistics

● Initiation Stage (3.24 - 3.29)

Task Name	complete status
Requirements analysis and confirmation	100 %
Project Charter Development	100%
Project Scope Definition	100%
Stakeholder Identification	100%
Resource Allocation	100%

The total Number of Tasks during Initiation Stage is 5

Tasks Statistics

● Planning Stage (3.30 - 4.6)

Task Name	complete status
Detailed Requirements Specification	100 %
Test Plan Development	100%
Risk Management Plan	100%
Model Selection	100%
Data Requirements Analysis	100%

The total Number of Tasks during Planning Stage is 5

Tasks Statistics

● Design Stage (4.2 - 4.22)

Task Name	complete status
System Architecture Design	100 %
AI Model Design	100%
Interface Design	100%
Database Design	100%
Core Function Design (UI+backend)	100%
.....	

The total Number of Tasks during Design Stage is 6

Tasks Statistics

● Development Stage (4.4 - 5.17)

Task Name	complete status
Development Environment Setup	100 %
Database Implementation	100%
Backend Service Development	100%
Core Function Development with Unit System	100%
Non-Core Function Development with Unit Test	100%
.....	

The total Number of Tasks during Development Stage is 11

Tasks Statistics

● Testing Stage (4.27 - 5.26)

Task Name	complete status
Core System Integration Testing	100 %
System Testing	100%
AI Model Validation	100%
User Acceptance Testing	100%

The total Number of Tasks during Tesing Stage is 4

Tasks Statistics

• Deployment&Launch Stage (5.27 - 6.3)

Task Name	complete status
Deployment Environment	100 %
System Deploy	100%
System Debugging	100%
Official Launch	100%

The total Number of Tasks during Deployment&Launch Stage is 4

Tasks Statistics

● Closure Stage (6.4 - 6.13)

Task Name	complete status
System Documentation Finalization	100 %
Project Documentation Archiving	100%
Lessons Learned Documentation	100%
Project Closure Report	100%

The total Number of Tasks during Closure Stage is 4

So the total number of tasks of our project is 39



Working hours Statistics

● Working hours of members

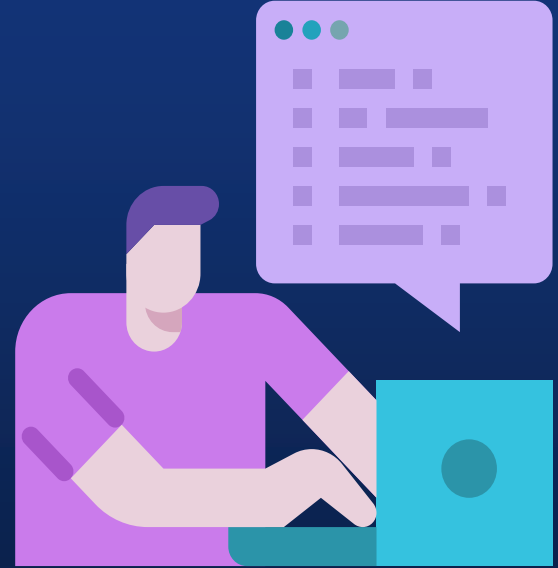
Member	First Biweekly	Second Biweekly	Third Biweekly	Fourth Biweekly	Fifth Biweekly	Total
Xuanhe Yang	30	31	32	30	30	153
Fubin Chen	22	36	36	32	30	156
Jingxiao Han	22	30	56	30	25	163
Tong Li	20	32	52	34	27	165

The working hours of our staff are relatively close and the division of labor is even

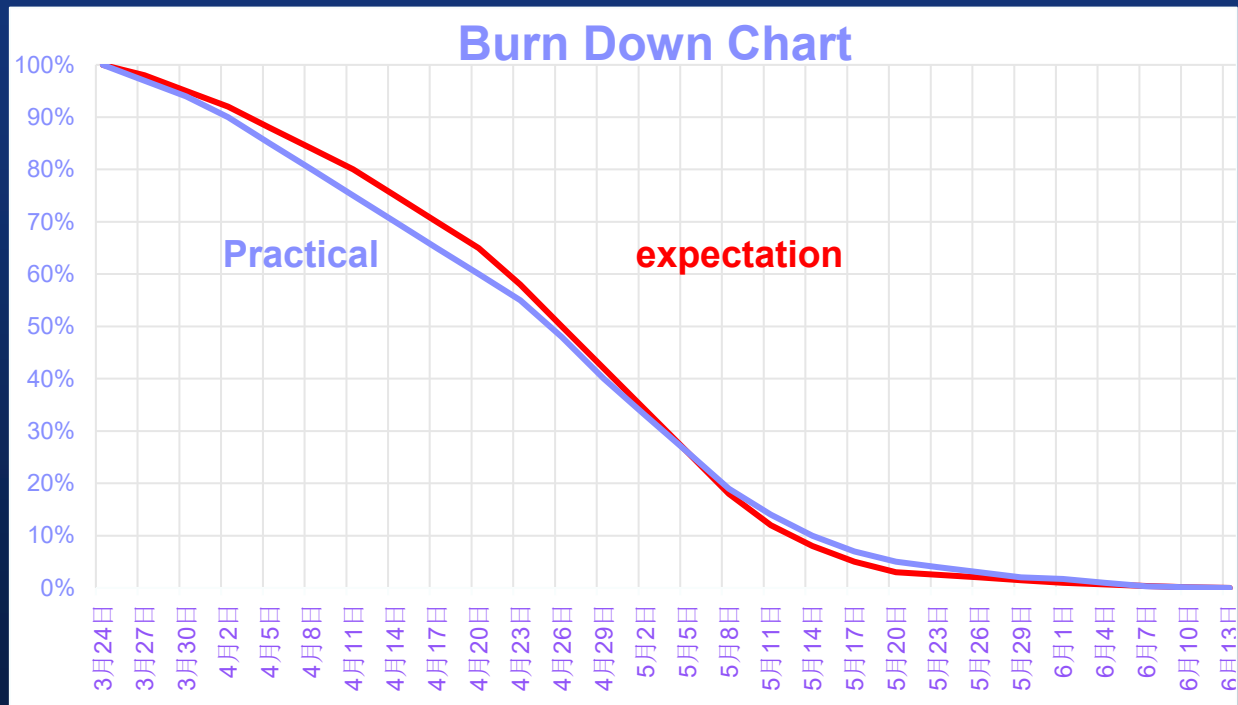


05

Compare between
plan &&
actual project



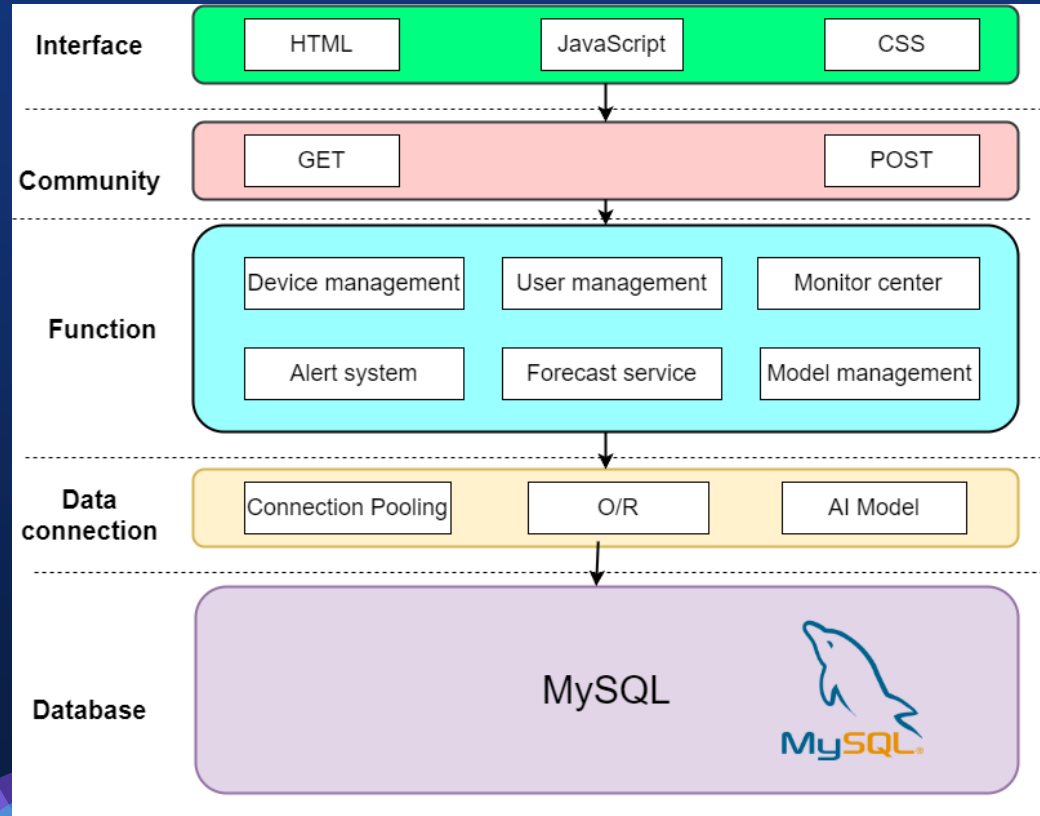
Comparison of completion status



- In the early stage, our progress was **faster** than planned because we made **preparations** for the project in advance
- In the test stage, our progress was **slower** because of the AI model. We must **adjusting the connection logic** between the back-end and the front-end. And **adjusted AI model**.

Finally, we completed the task on schedule !

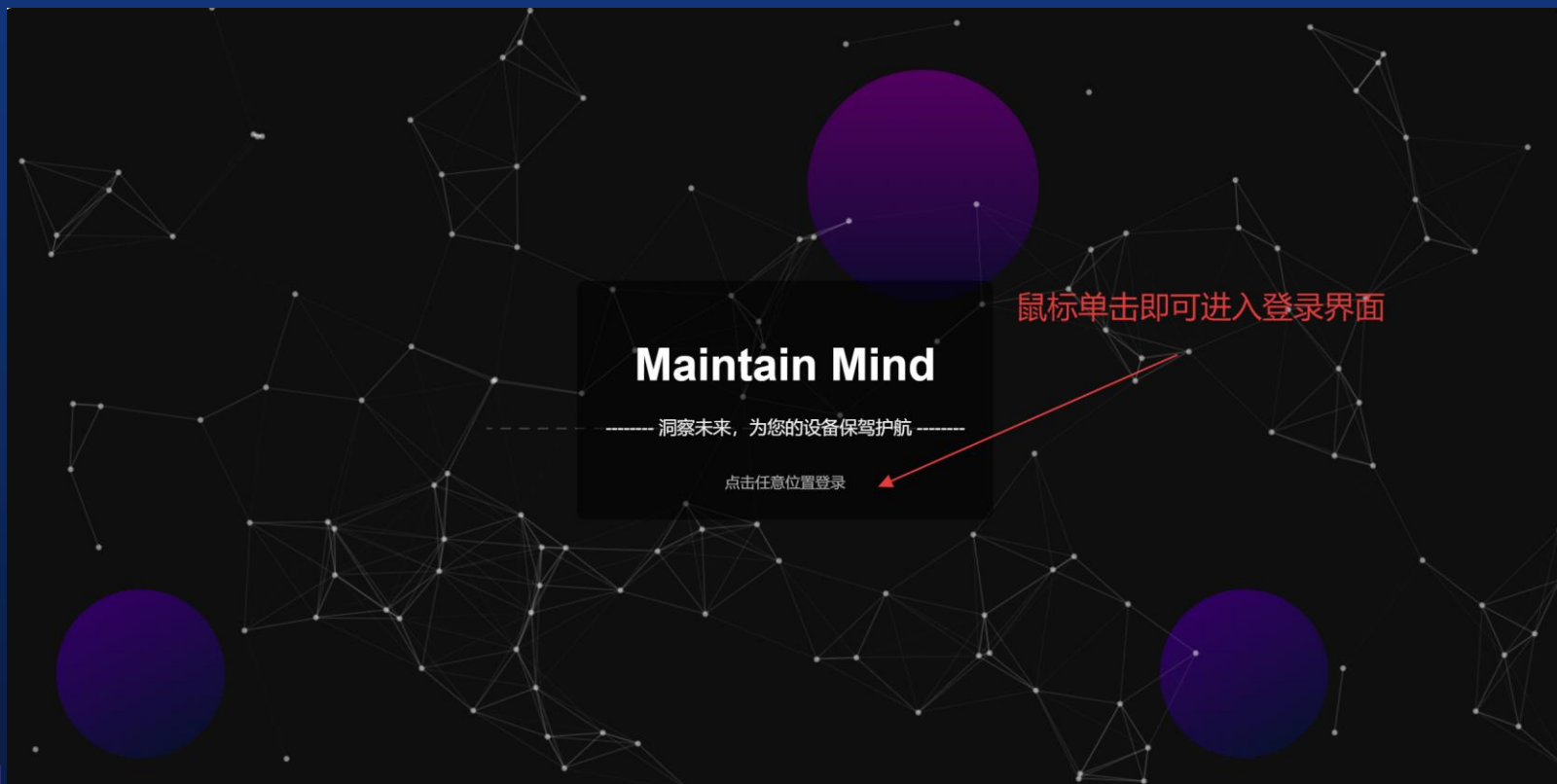
Comparison of Software implementation



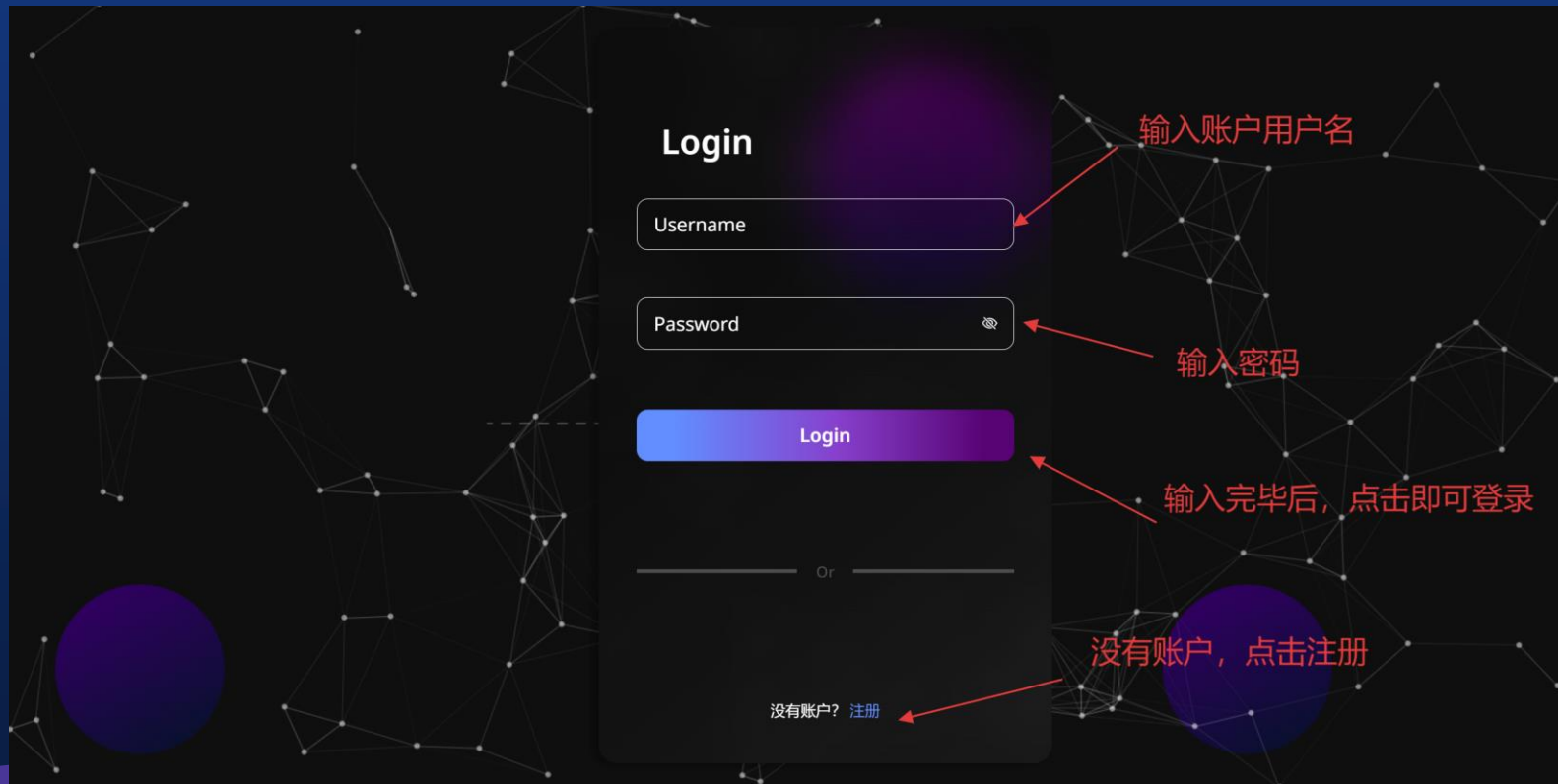
In the plan, we expect following functions :

- User Part
- Device management
- Device Monitor
- AI predicts Device health
- Alert &&Report

User Part Effect



User Part Effect



Device management Effect



Monitor Effect



Simulation Effect



Alert system Effect

 Tongji University

 设备中心

 监测中心

 数据模拟

 警报系统

 报表系统

搜索设备

批量确认

开始时间

结束时间

导出报表

点击该按钮导出预警设备excel表格

警报列表

共 18 条10条/页1前往1页

该页面主要显示出现预警的设备以及原因, 可以点击确认

Report system Effect





06

Size estimation



Size estimation

• IFPUG Method

ILF and EIF Complexity Analysis

ILF	DET	RET	Complexity	FP
User Management Data	14	2	Low	7
Equipment Management Data	18	3	Low	7
Alert Record Data	12	2	Low	7
Model Management Data	15	3	Low	7
Equipment Health Status Data	25	4	Average	10
Energy Consumption Data	10	2	Low	7
Predictive Analysis Data	22	3	Average	10
EIF	DET	RET	Complexity	FP
Country/Region Code Data	5	1	Low	5
Sensor Device Data Stream	35	4	Average	7

Size estimation

• EI EO EQ Complexity Analysis

EI	DET	FTR	Complexity	FP
User Registration	8	1	Low	3
User Login	3	1	Low	3
User Information Management	12	1	Low	3
Equipment Addition	15	2	Average	4
Equipment Info Modification	18	2	High	6
Equipment Deletion	3	2	Low	3
Alert Confirmation	5	1	Low	3
Alert Deletion	3	1	Low	3
Model Import	12	2	Average	4
Model Selection	6	2	Average	4
Simulation Parameter Input	20	3	High	6

EO	DET	FTR	Complexity	FP
Equipment Health Status Report	18	3	Average	5
Equipment Energy Report	12	2	Average	5
Alert Statistics Report	10	2	Average	5
Fault Trend Report	15	3	Average	5
Equipment Status Report	14	2	Average	5
Life Prediction Report	16	3	Average	5
Health Data Report	14	3	Average	5

EQ	DET	FTR	Complexity	FP
Equipment List Query	12	2	Average	4
Equipment Details Query	18	3	Average	4
Alert Record Query	10	2	Average	4
Historical Data Query	15	3	Average	4
Model List Query	8	1	Low	3
Model Information Query	12	2	Average	4

Size estimation

- **UFP (Unadjusted function point)**

$$\text{UFP} = \text{ILF} + \text{EIF} + \text{EI} + \text{EO} + \text{EQ} = 55 + 12 + 42 + 35 + 23 = 167$$

- **14 General System Characteristics**

General System Characteristic	Rating (0-5)
Data Communications	5
Distributed Data Processing	2
Performance	4
Heavily Used Configuration	3
Transaction Rate	3
Online Data Entry	5
End-User Efficiency	4
Online Update	4
Complex Processing	5
Reusability	3
Installation Ease	2
Operational Ease	3
Multiple Sites	2
Facilitate Change	3
Total	48

- **VAF(Value Adjustment Factor)**

$$\text{VAF} = 0.65 + 0.01 \times 48 = 1.13$$

- **Adjusted function points (FA)**

$$\text{FP} = \text{UFP} \times \text{VAF} = 167 \times 1.13 = 188.71$$



07

Cost estimation



Cost estimation

• Selecting an Appropriate Productivity Standard

 北京软件造价评估技术创新联盟		
使软件造价评估专业化		
Application Type	Complexity	Productivity (Person-Hours/FP)
Business Application System	Medium	6.83

• Converting to Calendar Time According to the Productivity Standard

Calculation of project development effort:

- $\text{Effort} = \text{Number of Function Points} \times \text{Productivity} = 189 \times 6.83 = 1,290.87 \text{ person-hours}$

Assuming 8 working hours per day and 22.5 working days per month:

- $\text{Monthly working hours} = 22.5 \times 8 = 180 \text{ person-hours}$
- $\text{Months required for project development} = 1,290.87 \div 180 = 7.17 \text{ months}$

Assuming 4 developers simultaneously participating in project development:

- $\text{Calendar time required for project development} = 7.17 \div 4 = 1.79 \text{ months, approximately 36 working days}$

Cost estimation

● Obtaining Software Cost

According to industry standard salary data, the average salaries for IT industry developers in Shanghai are 31,207 CNY/month.

Project development cost calculation:

- Labor cost = required effort $\times$ person monthly rate = $7.17 \times 31,207 = 223,100$ CNY
- Additional costs for hardware, software, training, etc. (estimated at 15% of labor cost):
- Other costs = $223,100 \times 15\% = 33,465$ CNY
- Total project cost = Labor cost + Other costs = $223,100 + 33,465 = 256,565$ CNY

Therefore, using the IFPUG method, the estimated total cost of the Proactive Shield project is approximately **256,565 CNY**.

● Verifying Results Using Another Function Point Method

To verify the above results, we used the IFPUG method for estimation:

Based on the function point count obtained using the IFPUG method from the previous experiment:

- Adjusted Function Points (FP): 196

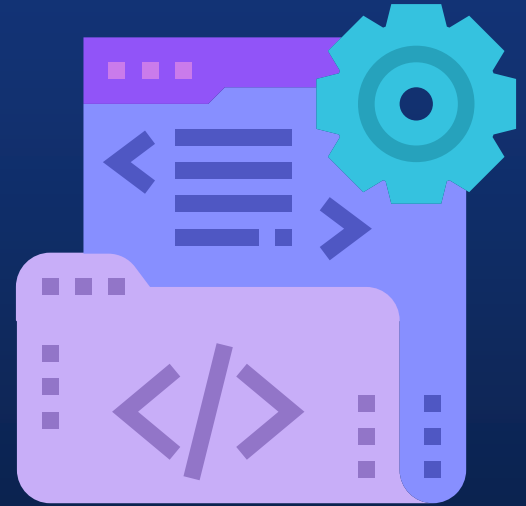
Using the same productivity standard:

- Effort = $196 \times 6.83 = 1,339.48$ person-hours
- Months = $1,339.48 \div 180 = 7.43$ months
- Calendar time for a 4-person team = $7.43 \div 4 = 1.86$ months
- Project cost = $7.43 \times 31,207 + 7.43 \times 31,207 \times 15\% = 232,000 + 34,800 = 266,800$ CNY

The difference between the two method's estimation results is: $(256,565 - 266,800) \div 266,800 \times 100\% = 3.84\%$, which is a very small difference, validating that our estimation results have high reliability.

08

Economic evaluation





Economic evaluation

● Economic Environment

- Development Period: 2.84 months (82 days, 58 working days)
- Team Size: 4 members (1 AI engineer + 1 frontend + 1 backend + 1 data engineer)
- Daily Working Hours: 16 person-hours/day
- Total Effort: 549 person-hours
- Actual Productivity: 2.56 person-hours/FP (based on 215 FP from Problem 1)
- Total Labor Cost: ¥268,380 (¥94,500/month $\times$ 2.84 months)

● Compliance with National Standards

This economic evaluation follows the guidelines from:

- GB/T 36964-2018: Software Engineering - Function Point Analysis
- National Development and Reform Commission Economic Evaluation Methods
- China Software Industry Association Cost Estimation Standards
- Ministry of Industry and Information Technology Software Pricing Guidelines

The discount rate (12%) aligns with NDRC recommended rates for high-tech software projects.



Economic evaluation

● Business Model

Positioned as a premium B2B SaaS platform for aerospace manufacturers, large industrial facilities, and maintenance service providers. Core revenue streams:

- Software Licensing: Annual subscription model for platform access
- Implementation Services: One-time setup and integration fees
- Premium Support: Enhanced technical support and customization

● Pricing Strategy

Market Benchmarking:

Benchmark	Price Range
IBM Maximo Asset Management	\$150-\$300/user/month
GE Predix Platform	\$50,000-\$200,000/year (enterprise)
Siemens MindSphere	\$30,000-\$150,000/year
SAP Asset Intelligence Network	\$100,000-\$500,000/year
Domestic Industrial IoT Platforms	¥50,000-¥300,000/year

- Basic License: **¥150,000**/installation/year
- Implementation Service: ¥100,000/project (one-time)
- Premium Support: ¥40,000/installation/year

Economic evaluation

• Cost Estimation (¥10,000)

Cost Item	Amount	Description
Development Labor	35.0	Extended development cycle and team costs
Marketing & Promotion	12.0	Market expansion and customer acquisition
Server Infrastructure	8.0	Enhanced cloud services and scalability
Software Licenses	3.5	Advanced development tools and libraries
Maintenance & Support	6.0	Comprehensive platform maintenance
Working Capital	4.0	Operational cash flow requirements
Total Cost	68.5	

Economic evaluation

• Benefit Estimation (¥10,000)

Year	Implementation	Basic License	Premium Support	Total Income
1	15.0	10.0	2.0	27.0
2	20.0	25.0	5.0	50.0
3	15.0	40.0	8.0	63.0
4	10.0	55.0	12.0	77.0
5	8.0	70.0	15.0	93.0

• Revenue Assumptions:

- Year 1 focuses on customer acquisition with 2 major aerospace manufacturers
- Year 2 expands to 3-4 clients including maintenance service providers
- Year 3+ achieves scale with 95% customer retention rate
- Premium support adoption increases from 25% initially to 35% eventually
- Pricing based on aerospace industry standards and competitive analysis

Economic evaluation

• Capital Usage Plan

Financial Analysis

• Basic Statement 1:

Investment Cash Flow Table (¥10,000)

Usage Category	Amount (¥10,000)	Percentage
R&D and Development	35.0	51.1%
Marketing & Sales	12.0	17.5%
Infrastructure	8.0	11.7%
Working Capital	4.0	5.8%
Software & Tools	3.5	5.1%
Maintenance & Support	6.0	8.8%
Total	68.5	100%

Operating Cost Breakdown:

- Operations & Maintenance: 6.0
- Additional operational expenses: 9.0
- Total OpEx: 15.0 (Year 1), scaling up annually

Year	Income	Investment	OpEx	Net Cash Flow
0	0	68.5	0	-68.5
1	27.0	0	27.48	-0.48
2	50.0	0	34.0	16.0
3	63.0	0	39.85	23.15
4	77.0	0	45.78	31.22
5	93.0	0	51.84	41.16

Economic evaluation

- **Basic Statement2: Project Equity Cash Flow Statement(¥10,000)**

Year	Cash Inflow	Cash Outflow	Net Cash Flow	Cumulative Cash Flow
0	68.5	68.5	-68.5	-68.5
1	27.0	27.48	-0.48	-68.98
2	50.0	34.0	16.0	-52.98
3	63.0	39.85	23.15	-29.83
4	77.0	45.78	31.22	1.39
5	93.0	51.84	41.16	42.55

- **Basic Statement3: Income and Profit Distribution Statement(¥10,000)**

Year	Operating Income	Total Cost	Gross Profit	Income Tax	Net Profit
1	27.0	50.32	-23.32	0	-23.32
2	50.0	24.7	25.3	6.33	18.97
3	63.0	29.48	33.52	8.38	25.14
4	77.0	34.32	42.68	10.67	32.01
5	93.0	39.28	53.72	13.43	40.29

Economic evaluation

● Auxiliary Statement1: Investment Estimation Table(¥10,000)

Item	Amount	Description & Data Source
Labor Cost	35.0	Extended development cycle and enhanced team
Marketing & Promotion	12.0	Customer acquisition and market expansion
Server Infrastructure	8.0	Enhanced cloud services and scalability
Software/Tools	3.5	Advanced development and analysis tools
Maintenance & Support	6.0	Comprehensive platform maintenance
Working Capital	4.0	Operational cash flow requirements
Total Investment	68.5	

● Auxiliary Statement2: Working Capital Estimation Table(¥10,000)

No.	Item / Year	Year 1	Year 2	Year 3	Year 4	Year 5
1	Current Assets	4.0	6.0	8.0	10.0	12.0
2	Current Liabilities	0	0	0	0	0
3	Working Capital	4.0	6.0	8.0	10.0	12.0
4	Annual Change	4.0	2.0	2.0	2.0	2.0

Economic evaluation

• Auxiliary Statement3: Total Cost Estimation Table (¥10,000)

No.	Item / Year	Year 1	Year 2	Year 3	Year 4	Year 5
1	Salaries & Benefits	12.0	15.0	18.0	21.0	24.0
2	Operations & Maintenance	6.0	8.0	10.0	12.0	14.0
3	Depreciation	0.7	0.7	0.7	0.7	0.7
4	Amortization	7.0	7.0	7.0	7.0	7.0
5	Taxes & Fees	1.78	3.30	4.15	5.08	6.14
6	Total Cost	27.48	34.0	39.85	45.78	51.84

• Auxiliary Statement 4:FixedAssetsDepreciationTable(¥10,000)

No.	Item / Year	Year 1	Year 2	Year 3	Year 4	Year 5
1	Equipment & Tools (Original Value)	3.5	0	0	0	0
1.1	New Fixed Assets Added	3.5	0	0	0	0
1.2	Annual Depreciation	0	0.7	0.7	0.7	0.7
2	Net Fixed Assets	3.5	2.8	2.1	1.4	0.7
3	Year-End Net Value	3.5	2.8	2.1	1.4	0.7

Economic evaluation

● Auxiliary Statement 5: Intangible Assets Amortization Table (¥10,000)

No.	Item / Year	Year 1	Year 2	Year 3	Year 4	Year 5
1	Software Development Cost	35.0	0	0	0	0
2	Annual Amortization	7.0	7.0	7.0	7.0	7.0
3	Year-End Net Value	28.0	21.0	14.0	7.0	0

● Auxiliary Statement 6: Value-Added Tax and Additional Estimation Table(¥10,000)

No.	Item / Year	Year 1	Year 2	Year 3	Year 4	Year 5
1	Operating Income	27.0	50.0	63.0	77.0	93.0
2	VAT (6%)	1.62	3.0	3.78	4.62	5.58
3	Urban Maintenance Tax (7%)	0.11	0.21	0.26	0.32	0.39
4	Education Surcharge (3%)	0.05	0.09	0.11	0.14	0.17
5	Total Taxes	1.78	3.30	4.15	5.08	6.14

Pre-financing Economic Evaluation and Feasibility Analysis

• Cash Flow Table

No.	Item / Year	Year 1	Year 2	Year 3	Year 4	Year 5
1	Operating Income	27.0	50.0	63.0	77.0	93.0
2	VAT (6%)	1.62	3.0	3.78	4.62	5.58
3	Urban Maintenance Tax (7%)	0.11	0.21	0.26	0.32	0.39
4	Education Surcharge (3%)	0.05	0.09	0.11	0.14	0.17
5	Total Taxes	1.78	3.30	4.15	5.08	6.14

• Key Financial Indicator Calculations

NPV:

NPV Calculation Formula:

$$NPV = \sum_{t=0}^n \frac{F_t}{(1+i)^t}$$

Where: $F_0 = -68.5$, $F_1 = 12.0$, $F_2 = 32.0$, $F_3 = 41.0$, $F_4 = 51.0$, $F_5 = 63.0$

The project discount rate is set at 12%, based on:

- Risk-free rate: 2.5% (10-year Chinese government bond yield)
- Market risk premium: 5.0% (average for software/IT industry)
- Project-specific risk premium: 4.5% (technology and market uncertainty)
- Total: 12%

Pre-financing Economic Evaluation and Feasibility Analysis

● Key Financial Indicator Calculations

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NPV Calculation Formula:

$$NPV = \sum_{t=0}^n \frac{F_t}{(1+i)^t}$$

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- Risk-free rate: 2.5% (10-year Chinese government bond yield)
- Market risk premium: 5.0% (average for software/IT industry)
- Project-specific risk premium: 4.5% (technology and market uncertainty)
- Total: 12%

Year	Net Cash Flow	Discount Factor	Present Value
0	-68.5	1.000	-68.5
1	12.0	0.893	10.7
2	32.0	0.797	25.5
3	41.0	0.712	29.2
4	51.0	0.636	32.4
5	63.0	0.567	35.7
			65.0

$$NPV = -68.5 + 10.7 + 25.5 + 29.2 + 32.4 + 35.7 = 65.0 \text{ (ten thousand yuan)}$$

Pre-financing Economic Evaluation and Feasibility Analysis

• Key Financial Indicator Calculations

IRR:

$$0 = \sum_{t=0}^n \frac{F_t}{(1 + IRR)^t}$$

By inputting the above cash flows into Excel's IRR function: IRR 25.8%

DPP:

$$DPP = t_0 + \frac{|Cumulative_{t_0}|}{Discounted_{t_0+1}}$$

Where:

- t_0 is the last year in which the cumulative net present value is negative
- $|Cumulative_{NPV_{t_0}}|$ is the absolute value of the cumulative net present value at the end of year t_0
- $Discounted_{NetCashFlow_{t_0+1}}$ is the discounted net cash flow for the next year

Year	Discounted Value	Cumulative NPV
0	-68.5	-68.5
1	10.7	-57.8
2	25.5	-32.3
3	29.2	-3.1
4	32.4	29.3

$$DPP = 3 + \frac{3.1}{32.4} = 3.1 \text{ years}$$

Final Feasibility Judgment

• Go/No-Go Decision

- NPV>0: (¥650,000)
- IRR >MARR: (25.8% >12%)
- DPP<4 years: (3.1 years)
- Risk tolerance: (Controllable)

Indicator	This Project	Industry Benchmark (Industrial Software)	Assessment
IRR	25.8%	15%-25%	Slightly above benchmark
DPP	3.1 years	2-4 years	Within benchmark range
NPV	¥0.65 million	>0 feasible	Positive and reasonable

Conclusion:

- The cumulative present value turns positive in the 2nd year (break-even point)
- Need to ensure stable cash flow for the first two years (recommended to reserve emergency funds)
- Even if client renewals or value-added service income fall short of expectations, overall risk remains controllable

Economic Benefit Evaluation (Post-Financing Analysis)

● Revenue Quality Analysis (After Financing):

Stage	Cash Flow Feature	Key Drivers
Investment (0-1 yrs)	Cumulative investment ¥380,000	Platform development, customer acquisition
Payback (1-2 yrs)	Annual net cash flow ¥300-380k	Customer renewals, service expansion
Profit (2+ yrs)	Net cash flow continues growing	Market penetration improvement

● Sensitivity Analysis:

Scenario	IRR Change	DPP Change	Risk Level
Revenue -20%	18.2%	3.8 years	Controllable
Cost +30%	19.5%	3.6 years	Controllable
Market delay 1 year	16.8%	4.2 years	Needs attention



THANKS